Reliable Agent-Driven Development

DETAILS	
Details	CNUG event
Date	22/04/2026
Speaker	Thomas Martinsen

Thomas Martinsen

Technical Evangelist @ Digital Engineering, Twoday

Microsoft Regional Director + Microsoft AI MVP

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The risks of using AI in development

So much for 'trust but verify': Nearly half of software developers don't check AI-generated code – and 38% say it's because it takes longer than reviewing code produced by colleagues

A concerning number of developers are failing to check AI-generated code, exposing enterprises to huge security threats

The Register®

AI coding tools make developers slower but they think they're faster, study finds

Predicted a 24% boost, but clocked a 19% drag

Thomas Claburn Fri 11 Jul 2025 // 22:41 UTC

Using AI might actually slow down experienced devs

News By Craig Hale published July 11, 2025

Experienced developers aren't getting any benefit from AI, report claims

Developers Lean on AI More, But Report Growing Doubts About Accuracy, Stack Overflow Survey Says

The Risks of Using AI in Software Development

Why AI Coding Tools Make Experienced Developers 19% Slower and How to Fix It

Oct 3, 2025 • Molisha Shah

Agent design and control in practice

Agent-driven development

AI features and capabilities

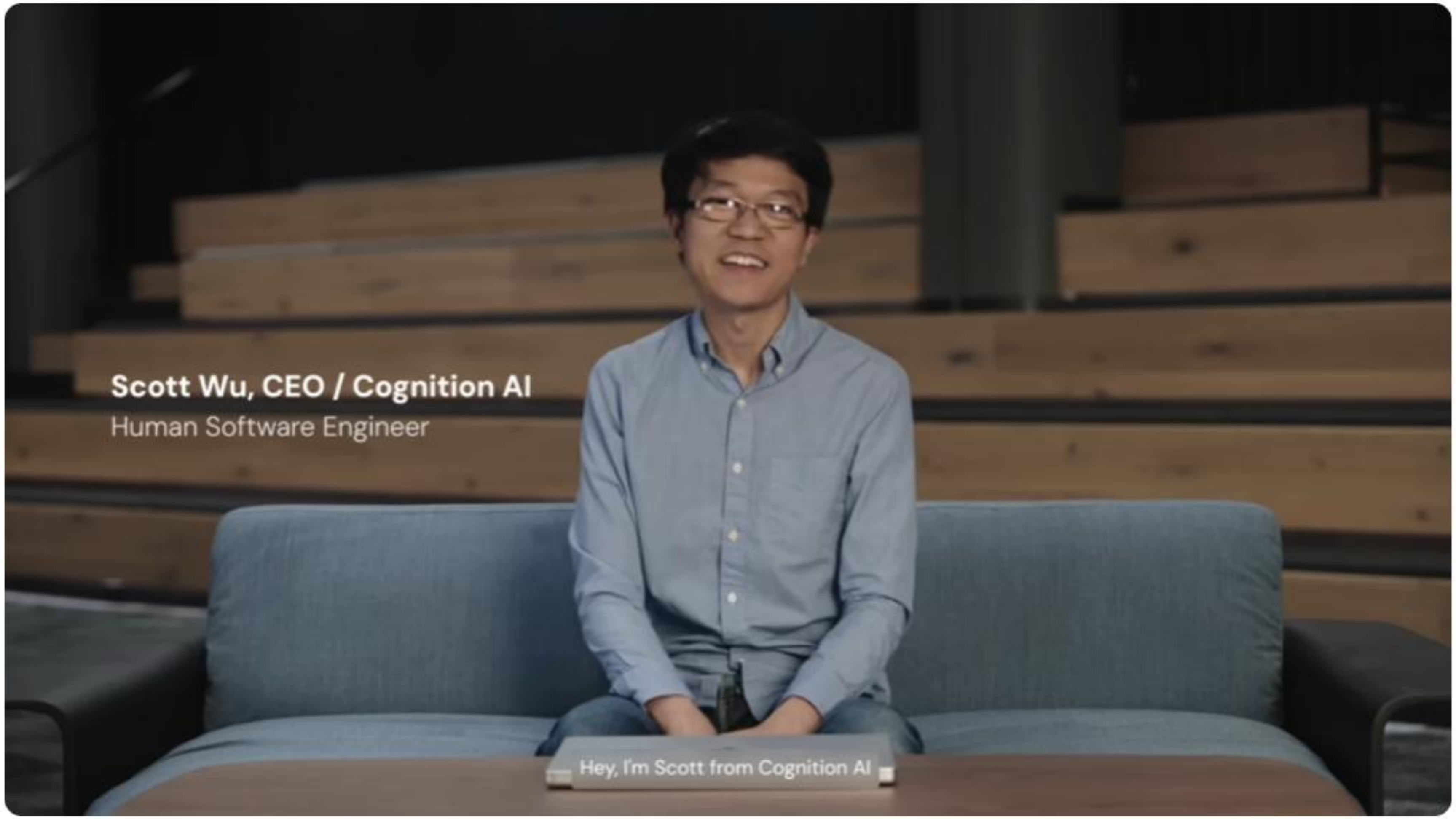
Agent-driven development is an approach to software engineering where AI agents participate actively in the development lifecycle rather than serving only as passive tools.

AI becomes a first-class architectural concern. It influences data strategy, security, governance, evaluation, and operational monitoring.

Scope - Skills - Knowledge - Instructions - Evaluation

Agentic Development


Scott Wu, CEO / Cognition AI
Human Software Engineer



Introducing Devin, the first AI software engineer

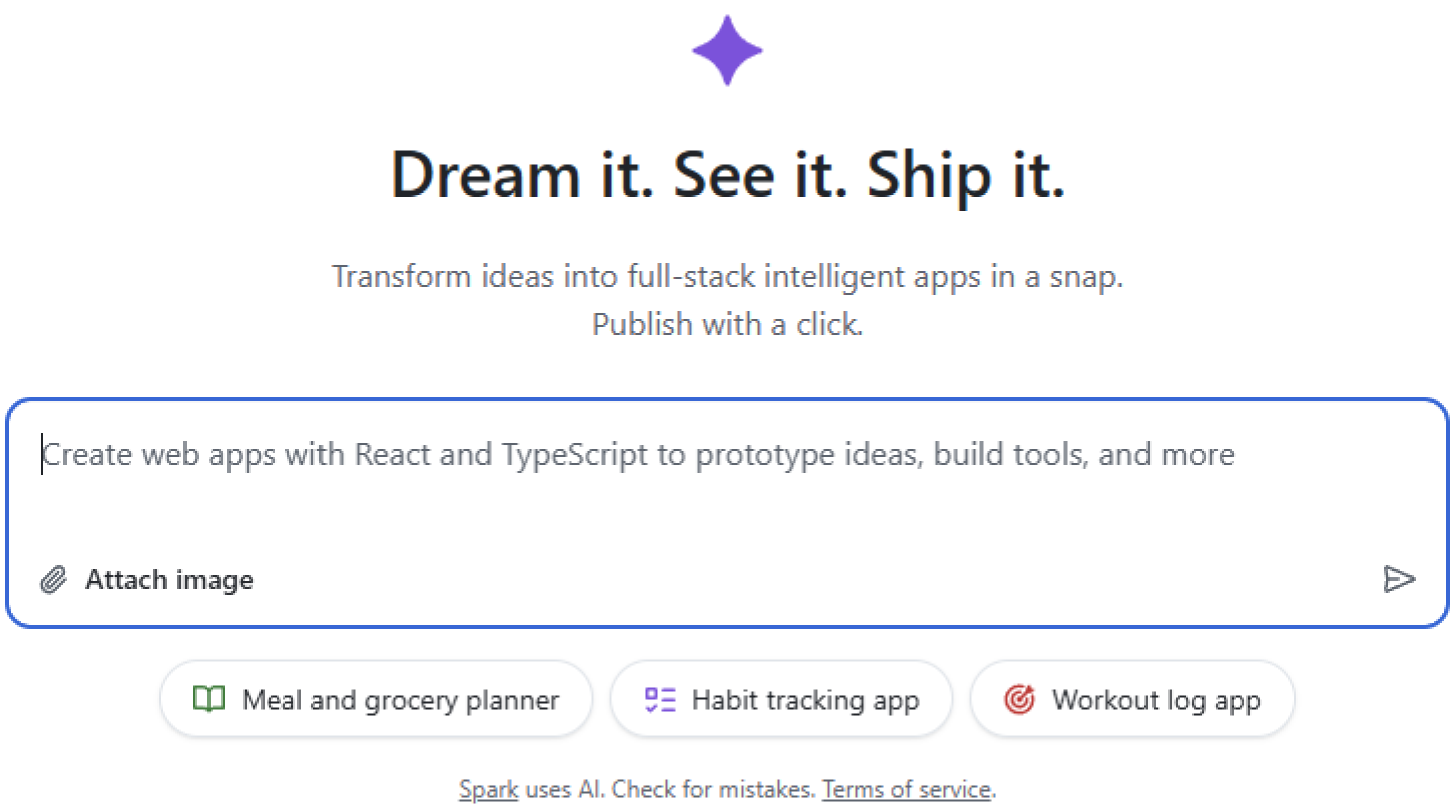
Cognition
28.4K subscribers
Subscribe

12K | Share | Save | Download

1.2M views 1 year ago
Meet Devin, the world's first fully autonomous AI software engineer.

Devin is a tireless, skilled teammate, equally ready to build alongside you or independently complete tasks for you to review. ...more

Vibe coding



https://github.com/spark

How development changes

MANUAL CODING

- Developer writes everything
- Manual scaffolding
- Hand-written tests
- Large code surface

AI-ASSISTED DEVELOPMENT

- Developer writes the code
- AI code suggestions
- Prompt-driven scaffolding
- AI-assisted refactoring

AGENT-DRIVEN DEVELOPMENT

- Developer defines intent
- Spec-driven planning
- Agent execution
- Human validation gates

Before

Manual coding

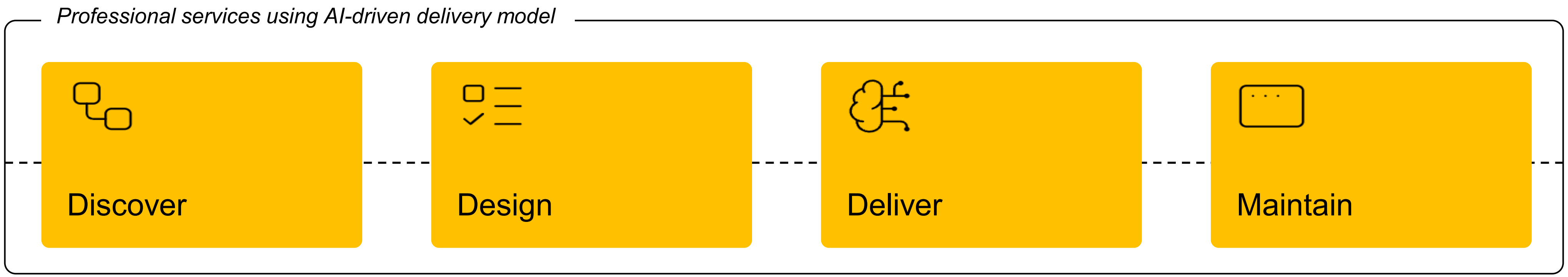
Producer → Intent designer + reviewer

Now

Agent-driven development

Delivery of advisory and professional services an using agent-driven approach

Roles
Innovation lead, Discovery facilitator, UX specialist, AI engineer, QA approver, Project orchestrator



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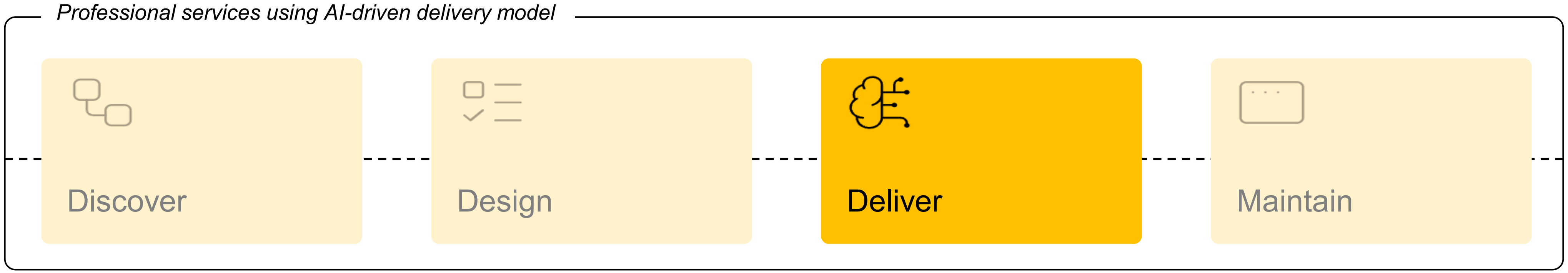

Deliver using agent-driven development

Structured, AI-orchestrated processes where the developer defines intent, specifications guide planning, agents execute tasks, and human validation gates ensure control and quality across the end-to-end flow. Designed to be predictable, reusable, team-scalable, and governed.

Discover

Design

Deliver

Maintain

DEVELOPMENT FLOW

Getting started

Enable agent-driven development structure and flow to software project.

Structuring knowledge

Describe infrastructure, audience, security model, tech stack, architecture.

Engineering primitives

Add agents, instructions, skills, and prompts to the solution.

Compile

Spec-driven development

Use spec engine to work on features and tasks and follow spec-driven principals.

Compile into standard formats

Compile knowledge and primitives into standard formatted artifacts.

Orchestrate and review changes

Use human orchestration and review, and AI test and review

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Discover | Design | **Deliver** | Maintain

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Demo: Structuring knowledge

KNOWLEDGE

- Architecture
- Audience
- Roles
- UX/UI
- Security
- Tech stack

Describe infrastructure, audience, security model, tech stack, architecture.

.github/copilot-instructions.md

openspec/config.yaml

ch Stack

Backend

- **.NET 10** — runtime and framework target
- **ASP.NET Core 10** — web host
- **FastEndpoints** — HTTP endpoint handler framework (replaces Controllers)
- **Entity Framework Core + Npgsql** — ORM + PostgreSQL driver
- **OpenAI SDK v2.x** (`OpenAI` NuGet) — OpenAI chat interactions
- **Azure AI Projects SDK** (`Azure.AI.Projects`) — Microsoft Foundry agent/chat/flow interactions
- **Model Context Protocol** (`ModelContextProtocol` NuGet) — MCP server tools
- **.NET Aspire 13.1** — orchestration and service composition
- **xUnit** — unit and integration tests
- **System.Text.Json** — serialization (Newtonsoft.Json is banned)

Frontend

- **React 19** — UI framework
- **TypeScript 5** — language
- **Vite 7** — build tooling and dev server
- **TanStack Query 5** — server state management
- **React Router 7** — client-side routing
- **Shadcn/UI + Tailwind CSS** — component library and styling
- **React Hook Form + Zod** — form handling and runtime validation
- **VS Code Codicons** — icon font for activity bar glyphs

Infrastructure / Platform

- **Azure Database for PostgreSQL Flexible Server** — primary database
- **Azure Cache for Redis** — caching layer
- **Azure Blob Storage** — file/content storage
- **Azure Key Vault** — secrets management
- **Microsoft Entra ID** — authentication
- **Azure Container Apps** — container hosting
- **Azure Monitor + Application Insights** — telemetry and monitoring

Demo: Engineering primitives

PRIMITIVES

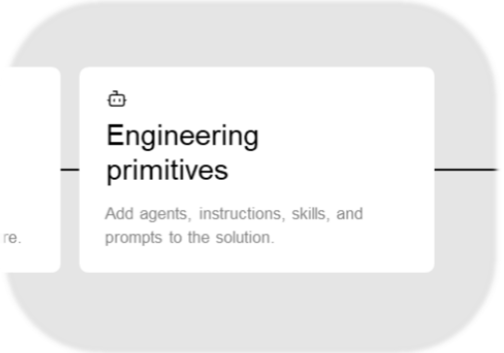
Agents | Instructions | Skills
Prompts | Plug-ins | Hooks

Add agents, instructions, skills, and prompts to the solution.

.github/agents/*
.github/instructions/*
.github/skills/*

Agents

Name	Filename	Description
Architect **[exists]**	`architect.agent.md`	Defines end-to-end architecture and contracts for the platform: API design, layer boundaries, cross-cutting concerns, security approach, and deployment topology.
Backend Developer **[exists]**	`developer-backend-dotnet.agent.md`	Implements the .NET 10 backend using Clean Architecture, Aspire, FastEndpoints, EF Core, and PostgreSQL. Handles REST API endpoints, auth validation, and unit/HTTP tests.
Frontend React Developer **[exists]**	`developer-frontend-react.agent.md`	Implements the React 19 + TypeScript 5 + Vite 7 frontend. Delivers UI flows, TanStack Query integration, React Router navigation, Shadcn/UI components, and Tailwind CSS styling.
Reviewer **[exists]**	`reviewer.agent.md`	Reviews specs, API contracts, implementation code, and documentation for correctness, architecture alignment, and governance conformance. Produces structured, actionable feedback.
Security Review **[exists]**	`security-review.agent.md`	On-demand security reviewer. Scans source code, configuration, infrastructure, and CI/CD workflows for OWASP Top 10 weaknesses. Provides prioritized remediation guidance.



agent = role
instructions = guidance and norms
prompt = current task
skill = reusable capability
plugin = installable package of capabilities
hook = event-driven automation around execution

Agentic building blocks

Primitives are the core configurable elements that developers iteratively refine to ensure reliable outcomes.

Engineering primitives

Add agents, instructions, skills, and prompts to the solution.

Agents

Reusable agents that orchestrate the role, permissions and operating boundaries.

Instructions

Scoped rules and the standards that the agent must follow and in which context.

Skills

Capabilities the agent can use within specific areas and context windows.

Orchestration

Use spec engine to work on features and tasks and follow spec-driven principals.

Specifications

Source of truth covering intent, constraints and acceptance criteria.

Memory

Decisions, notes and state maintained and carried forward.

agent = role
instructions = guidance and norms
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Spec-driven development

Follow spec-driven principals: Change => Implement => Archive.
Use human orchestration and review, and AI test and review

SPEC-DRIVEN DEVELOPEMENT

Feature specification

Planning and context

Implementation

Validation and test

Integration

Approve and review **specification**

Approve and review **architecture**

Approve and review **code**

Approve and review **test**

Demo: Spec-driven development

Install OpenSpec

npm install -g @fission-ai/openspec@latest

Initialize OpenSpec in project

openspec init

Use AI to fill OpenSpec config

/openspec/config.yaml

* https://github.com/Fission-AI/OpenSpec/

Demo: Spec-driven development

New branch

Create proposal for change request

/opsx:propose

Implement change request

/opsx:apply

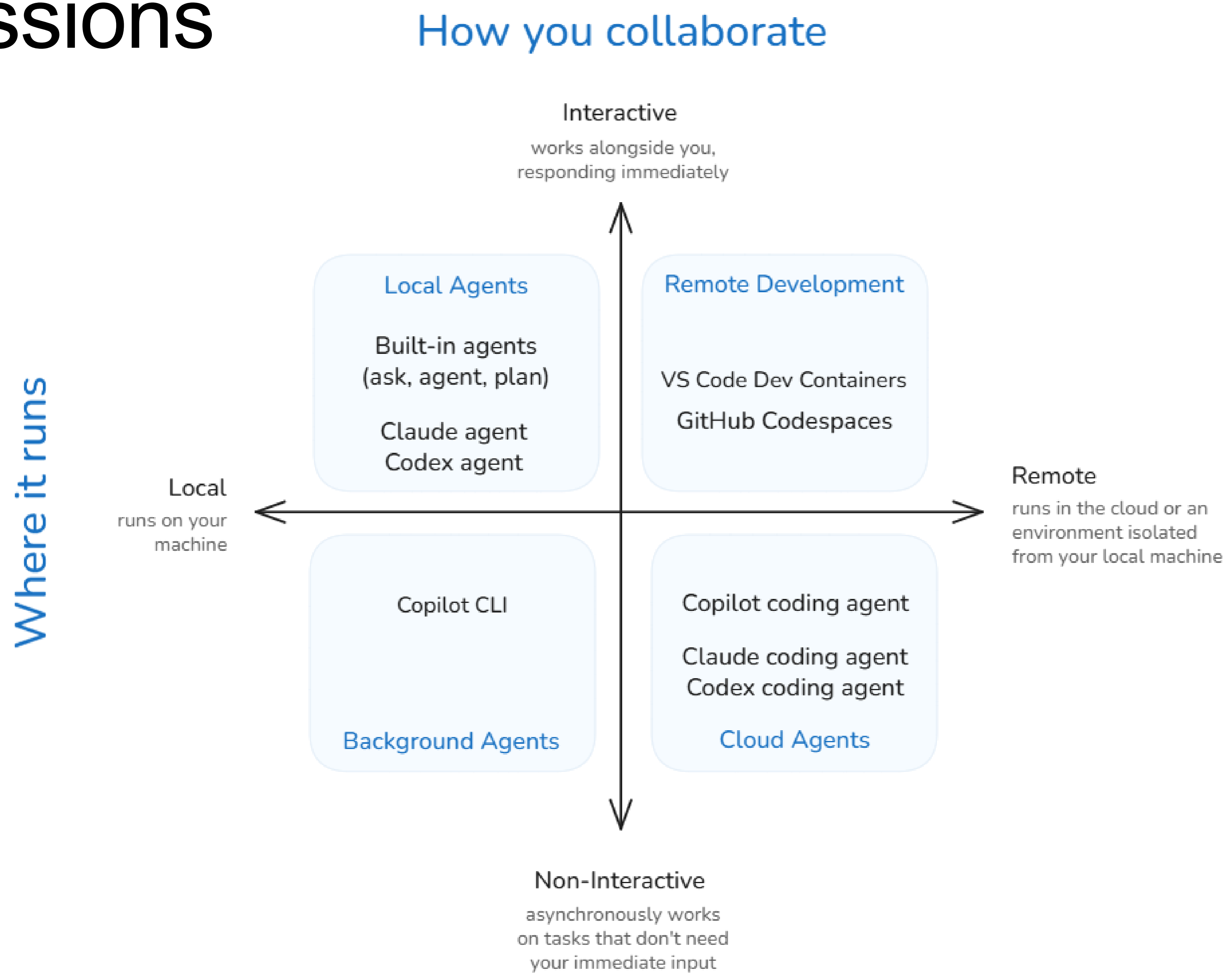
Archive change request

/opsx:archive

Pull Request

* https://github.com/Fission-AI/OpenSpec/

Agent sessions



Stay on top of your AI tools and resources

AI ORCHESTRATION

Visual Studio Code

GitHub Copilot* / Claude Code / Codex

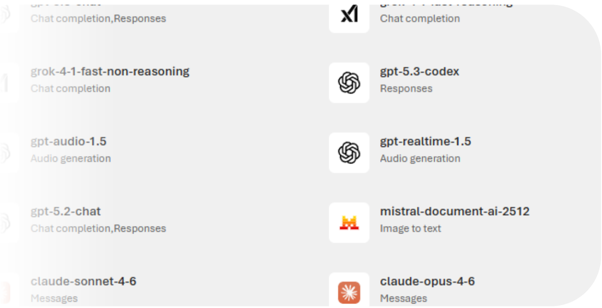
OpenSpec / GitHub SpecKit

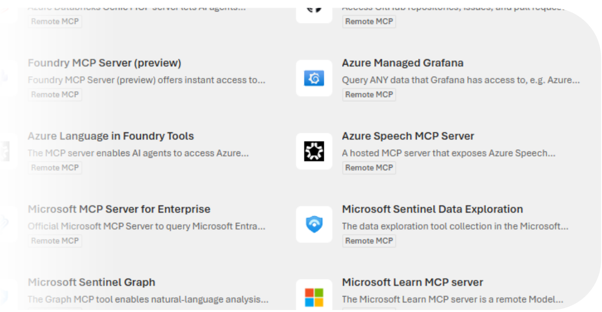
SERVERS AND MODELS

MCP servers (Azure, Learn, GitHub)

Models for planning

Models for implementation





* Business (300 PRPM) or Enterprise (1500 PRPM)

Predictable, reusable, and scale across teams

Development time decrease with
+50%

Quality and security up with
+20%

Improve standardization
>50%

Happy developers
+20%

Agentic Engineering in Twoday

We enable predictable, reusable, and team-scalable software delivery by orchestrating AI across workflows - replacing ad hoc prompting with structured specifications and governed execution

Agentic Engineering

Move from AI experimentation to scalable engineering impact

Starter Kit

Including playbook, workflow templates, pattern library and governance guidance

Catalogues

Catalogue of reusable primitives such as instructions, skills and review patterns

Events

Train teams in agentic development through hands-on masterclasses etc.

Enablement

Equip champions to support adoption and internal knowledge transfer

Questions

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